TEST REPORT

Application No.:	GZCR2406000623AT
Applicant:	CUBEPILOT PTY. LTD.
Address of Applicant:	Unit 2, 3 Dyson Court, Breakwater, Geelong, Victoria, 3219, Australia
Manufacturer:	CUBEPILOT PTY. LTD.
Address of Manufacturer:	Unit 2, 3 Dyson Court, Breakwater, Geelong, Victoria, 3219, Australia
Equipment Under Test (EUT):	
EUT Name:	Here4 Multiband RTK GNSS
Model No.:	HX4-06248
Standard(s) :	47 CFR Part 15, Subpart B
Date of Receipt:	2023-07-10
Date of Test:	2023-07-12 to 2023-07-18
Date of Issue:	2024-08-28
Test Result:	**Pass***

* In the configuration tested, the EUT complied with the standards specified above.

Jerry Chan
Manager

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Revision Record			
Version	**Report No.**	**Date**	**Remark**
01	GZCR230700069301	2023-08-24	Original
C02	GZCR230700069301C02	2024-08-28	Amendment report: Changed the information of Applicant and Manufacturer

Authorized for issue by:				
		Jim Li **Jim Li/Project Engineer**		
		Vico Cui **Vico Cui/Reviewer**		

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2 Test Summary

Emission Part				
Item	**Standard**	**Method**	**Requirement**	**Result**
Conducted Emissions at Mains Terminals (150kHz-30MHz)	47 CFR Part 15, Subpart B	ANSI C63.4:2014	15.107(a); Class B	Pass
Radiated Emissions (30MHz-1GHz)		ANSI C63.4:2014	15.109(a); Class B	Pass
Radiated Emissions (Above 1GHz)		ANSI C63.4:2014	15.109(g); Class B	Pass

Note:

E.U.T./EUT means Equipment Under Test.

Pass means the test result passed the test standard requirement, please find the detailed decision rule in the report relative section.

Remark for report GZCR230700069301C02:

This report GZCR230700069301C02 was based on original report GZCR230700069301, only changed the information of Applicant and Manufacturer.

Test data in original report GZCR230700069301 was kept in this report GZCR230700069301C02.

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4 General Information

4.1 Details of E.U.T.

Power supply: DC 5.3V(Powered by The Cube Orange+ Standard Set)

Cable(s): CAN cable, 0.2m, unshielded.

4.2 Description of Support Units

Description	Manufacturer	Model No.	Serial No.
Note Book Computer	LENOVO	ThinkPad T490	PF1D1MVJ
The Cube Orange+ Standard Set	HEXAERO PTE. LTD.	HX4-06222	/

4.3 Measurement Uncertainty

Test Item	Measurement Uncertainty
Conducted Emissions at Mains Terminals (150kHz-30MHz)	2.76dB (150kHz to 30MHz)
Radiated Emissions (30MHz-1GHz)	5.00dB (30MHz-1GHz):3m; 4.38dB (30MHz-1GHz):10m
Radiated Emissions (Above 1GHz)	5.12dB (1GHz-6GHz); 5.38dB (6GHz-18GHz); 5.61dB (18GHz-40GHz)

4.4 Test Location

All tests were performed at:

SGS-CSTC Standards Technical Services Co., Ltd., Guangzhou Branch EMC Laboratory,

No.198, Kezhu Road, Science City, Economic & Technological Development Area, Guangzhou, Guangdong, China 510663

Tel: +86 20 82155555

No tests were sub-contracted.

SGS-CSTC Standards Technical Services Co., Ltd. Guangzhou Branch Testing Center EEC Laboratory.

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4.5 Test Facility

The test facility is recognized, certified, or accredited by the following organizations:

- **ACMA**

SGS-CSTC Standards Technical Services Co., Ltd., EMC Laboratory can also perform testing for the Australian/New Zealand Regulatory Compliance Mark (RCM).

- **SGS UK(Certificate No.: 32), SGS-TUV SAARLAND and SGS-FIMKO**

Have approved SGS-CSTC Standards Technical Services Co., Ltd., EMC Laboratory as a supplier of EMC TESTING SERVICES and SAFETY TESTING SERVICES.

- **FCC Recognized Accredited Test Firm(Registration No.: 486818)**

SGS-CSTC Standards Technical Services Co., Ltd., EMC Laboratory has been accredited and fully described in a report filed with the (FCC) Federal Communications Commission. The acceptance letter from the FCC is maintained in our files. Designation Number: CN5016, Test Firm Registration Number: 486818.

- **ISED (Registration No.: 4620B, CAB identifier: CN0052)**

SGS-CSTC Standards Technical Services Co., Ltd., has been registered by Innovation Science and Economic Development Canada for Wireless Device Testing laboratories to test to Canadian radio equipment requirements. Registration No. 4620B, CAB identifier: CN0052.

- **VCCI (Registration No.: R-12460, C-12584, G-20107 and T-11179)**

The 10m Semi-anechoic chamber, 966 Anechoic Chamber and Shielded Room of SGS-CSTC Standards Technical Services Co., Ltd. have been registered in accordance with the Regulations for Voluntary Control Measures with Registration No.: R-12460, C-12584, G-20107 and T-11179 respectively.

- **CBTL (Lab Code: TL129)**

SGS-CSTC Standards Technical Services Co., Ltd., E&E Laboratory has been assessed and fully comply with the requirements of ISO/IEC 17025:2017, the Basic Rules, IECEE 01 and Rules of procedure IECEE 02, and the relevant IECEE CB-Scheme Operational documents.

4.6 Deviation from Standards

None

4.7 Abnormalities from Standard Conditions

None

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5 Equipment List

Conducted Emissions at Mains Terminals (150kHz-30MHz)					
Equipment	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal Date**	**Cal Due Date**
Coaxial Cable	HangTianXing	2m	EMC0107	2022-08-24	2023-08-23
Shielding Room	ChangZhou ZhongYu	8m x 3m x 3.8m	EMC0306	2022-10-16	2025-10-15
Two-Line V-Network-GZ	Rohde & Schwarz	ENV216	EMC2135	2022-09-09	2023-09-08
EMI Test Receiver (9kHz-3.6GHz)	Rohde & Schwarz	ESR3	EMC2221	2023-05-19	2024-05-18
Test Software E3r	Audix	Ver.6.11812	GZE100-77	N/A	N/A

Radiated Emissions (30MHz-1GHz)					
Equipment	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal Date**	**Cal Due Date**
10m Semi-Anechoic Chamber	ETS	N/A	EMC0530	2022-10-16	2025-10-15
Chamber cable	HangTianXing	N/A	EMC0542	2022-08-24	2023-08-23
Amplifier (9kHz-1.3GHz)	HP	8447F	EMC2065	2023-06-14	2024-06-13
EMI Test Receiver (1Hz-8GHz)	Rohde & Schwarz	ESW8	EMC2220	2023-05-19	2024-05-18
Test Software E3	Audix	Ver.6.120110a	GZE100-61	N/A	N/A
Trilog Broadband Antenna (25MHz-1GHz)	SCHWARZBECK MESS-ELEKTRONIK	VULB 9168	EMC2174	2022-06-19	2025-06-18

Radiated Emissions (Above 1GHz)					
Equipment	**Manufacturer**	**Model No.**	**Inventory No.**	**Cal Date**	**Cal Due Date**
1GHz-26.5 GHz Pre-Amplifier	Agilent	8449B	EMC0521	2022-12-16	2023-12-15
EMI Test Receiver (10Hz-26.5GHz)	Rohde & Schwarz	ESIB26	EMC0522	2022-12-16	2023-12-15
Chamber cable (Above 1GHz)	Scoflex	KMKM-8.0m	EMC0545	2022-08-24	2024-08-23
Horn Antenna (1GHz-18GHz)	SCHWARZBECK MESS-ELEKTRONIK	BBHA 9120D	EMC2026	2022-09-23	2025-09-22
966 Anechoic Chamber	C.R.T	9m x 6m x 6m	EMC2142	2020-12-20	2023-12-19
Test Software E3	Audix	Ver.6.120110a	GZE100-61	N/A	N/A

General used equipment					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
DMM	Fluke	73	EMC0006	2023-06-11	2024-06-10

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6 Emission Test Results

6.1 Conducted Emissions at Mains Terminals (150kHz-30MHz)

Test Requirement: 47 CFR Part 15, Subpart B

Test Method: ANSI C63.4:2014

Limit:

0.15MHz-0.5MHz: 66dB(μV)-56dB(μV) quasi-peak, 56dB(μV)-46dB(μV) average

0.5MHz-5MHz: 56dB(μV) quasi-peak, 46dB(μV) average

5MHz-30MHz: 60dB(μV) quasi-peak, 50dB(μV) average

Detector: Peak for pre-scan (9kHz resolution bandwidth) 0.15M to 30MHz

6.1.1 E.U.T. Operation

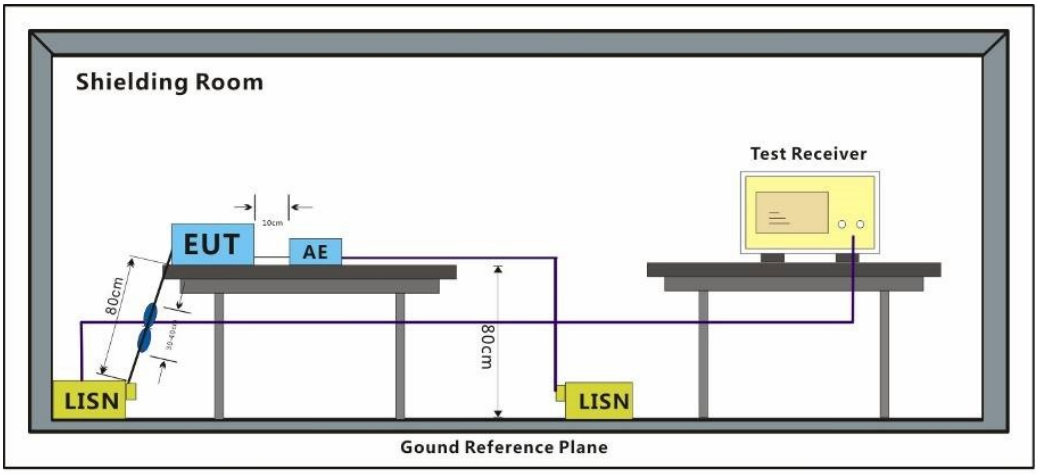
Operating Environment:

Temperature: 21.3 °C Humidity: 54.5 % RH Atmospheric Pressure: 1006 mbar

6.1.2 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Operation(GNSS):Keep the EUT power on and GPS connection through tools.

6.1.3 Test Setup Diagram



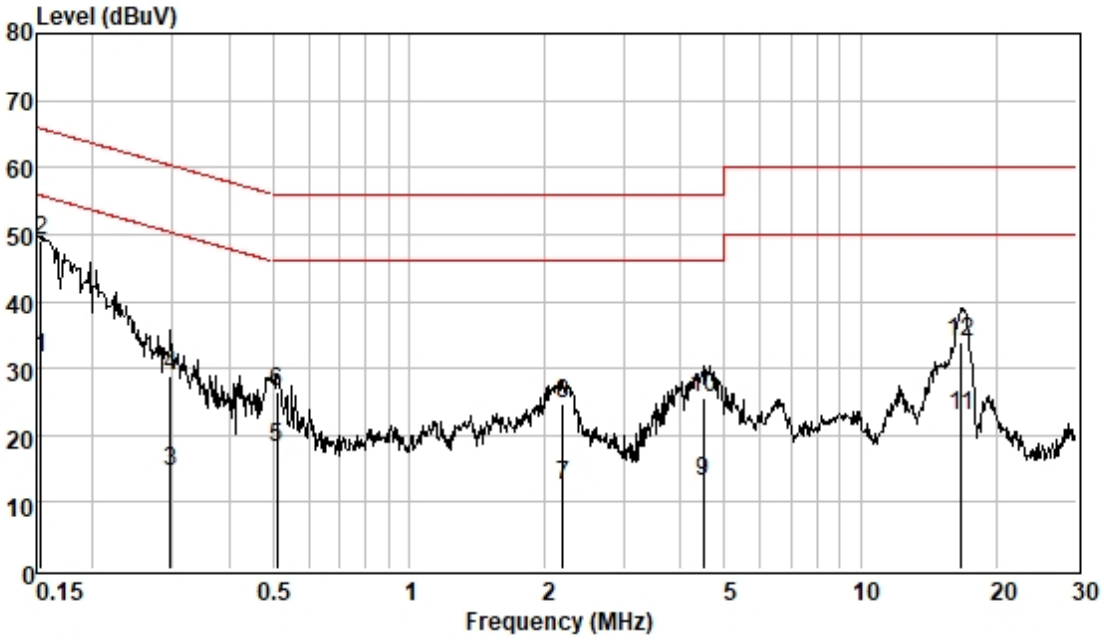
6.1.4 Measurement Procedure and Data

An initial pre-scan was performed with peak detector.Quasi-Peak or Average measurement were performed at the frequencies with maximized peak emission were detected.

Remark: Level= Read Level+ Cable Loss+ LISN Factor

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Test Mode: 00; Line: Live line

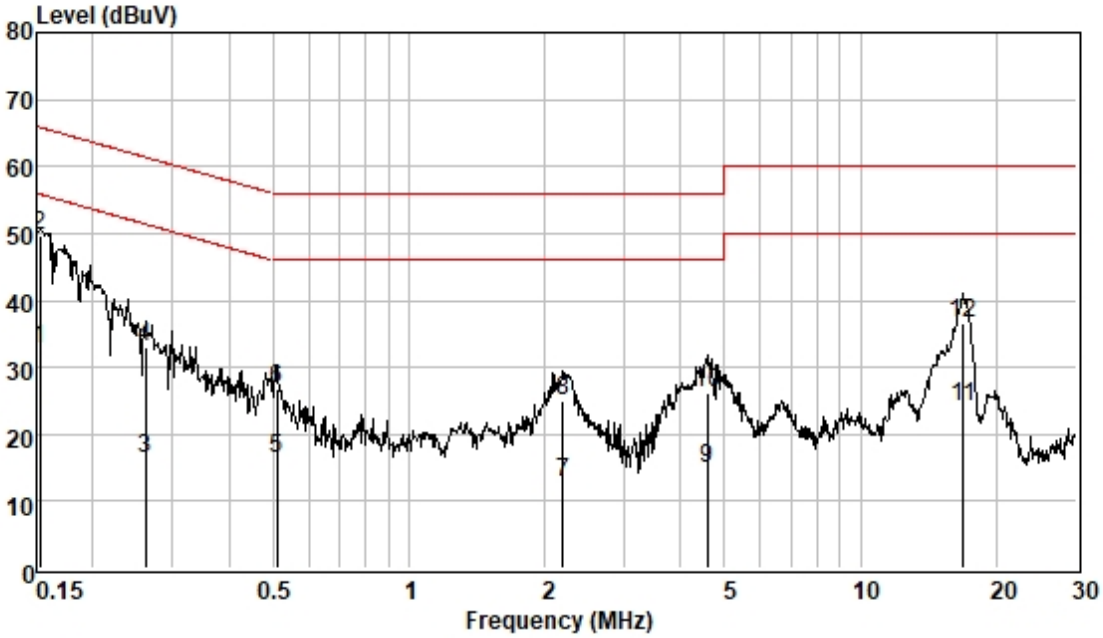


Pol :LINE
Mode :
Model :
Power :

	Frequenc MHz	Read Level dBuV	Cable Loss dB	LISN Factor dB	Measured Level dBuV	Limit Line dBuV	Over Limit dB	Remark
1	0.153	21.71	0.06	9.61	31.38	55.82	-24.44	Average
2	0.153	39.32	0.06	9.61	48.99	65.82	-16.83	QP
3	0.297	4.86	0.06	9.62	14.54	50.32	-35.78	Average
4	0.297	19.25	0.06	9.62	28.93	60.32	-31.39	QP
5	0.510	8.56	0.07	9.59	18.22	46.00	-27.78	Average
6	0.510	16.89	0.07	9.59	26.55	56.00	-29.45	QP
7	2.190	2.69	0.12	9.62	12.43	46.00	-33.57	Average
8	2.190	15.08	0.12	9.62	24.82	56.00	-31.18	QP
9	4.478	3.35	0.17	9.64	13.16	46.00	-32.84	Average
10	4.478	15.70	0.17	9.64	25.51	56.00	-30.49	QP
11	16.661	12.82	0.32	9.69	22.83	50.00	-27.17	Average
12	16.661	23.81	0.32	9.69	33.82	60.00	-26.18	QP

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Test Mode: 00; Line: Neutral Line



Pol :NEUTRAL
Mode :
Model :
Power :

	Frequenc MHz	Read Level dBuV	Cable Loss dB	LISN Factor dB	Measured Level dBuV	Limit Line dBuV	Over Limit dB	Remark
1	0.152	23.09	0.06	9.63	32.78	55.87	-23.09	Average
2	0.152	40.06	0.06	9.63	49.75	65.87	-16.12	QP
3	0.262	6.68	0.06	9.60	16.34	51.38	-35.04	Average
4	0.262	23.24	0.06	9.60	32.90	61.38	-28.48	QP
5	0.510	6.72	0.07	9.60	16.39	46.00	-29.61	Average
6	0.510	17.23	0.07	9.60	26.90	56.00	-29.10	QP
7	2.190	2.94	0.12	9.61	12.67	46.00	-33.33	Average
8	2.190	15.11	0.12	9.61	24.84	56.00	-31.16	QP
9	4.574	4.92	0.17	9.65	14.74	46.00	-31.26	Average
10	4.574	16.38	0.17	9.65	26.20	56.00	-29.80	QP
11	16.839	14.09	0.32	9.82	24.23	50.00	-25.77	Average
12	16.839	26.29	0.32	9.82	36.43	60.00	-23.57	QP

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6.2 Radiated Emissions (30MHz-1GHz)

Test Requirement:	47 CFR Part 15, Subpart B
Test Method:	ANSI C63.4:2014
Limit:	Class B
Test Distance:	10m
30MHz -88MHz	29.54 (dBμV/m) quasi-peak
88MHz-216MHz	33.06 (dBμV/m) quasi-peak
216MHz-960MHz	35.56 (dBμV/m) quasi-peak
960MHz-1000MHz	43.52 (dBμV/m) quasi-peak
Detector:	Peak for pre-scan (120kHz resolution bandwidth) 30MHz to1000MHz

6.2.1 E.U.T. Operation

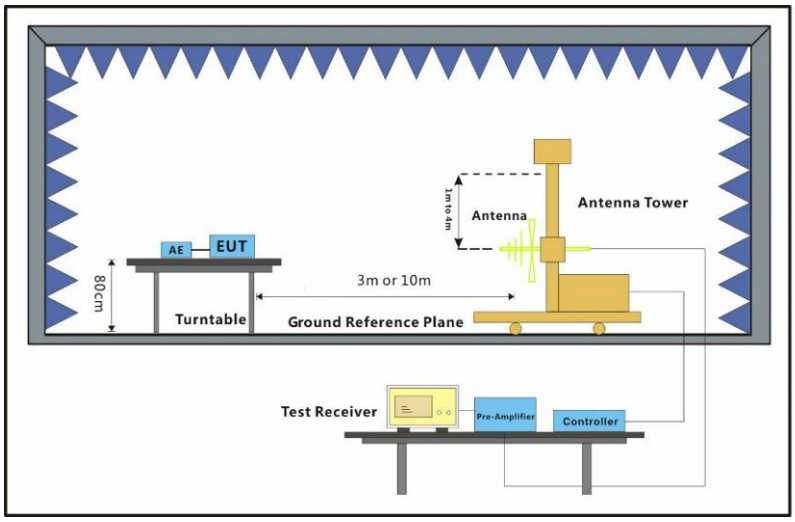
Operating Environment:

Temperature: 23.6 °C Humidity: 56.3 % RH Atmospheric Pressure: 1006 mbar

6.2.2 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Operation(GNSS):Keep the EUT power on and GPS connection through tools.
Pre-scan	01	Idle_Keep the EUT standby.

6.2.3 Test Setup Diagram



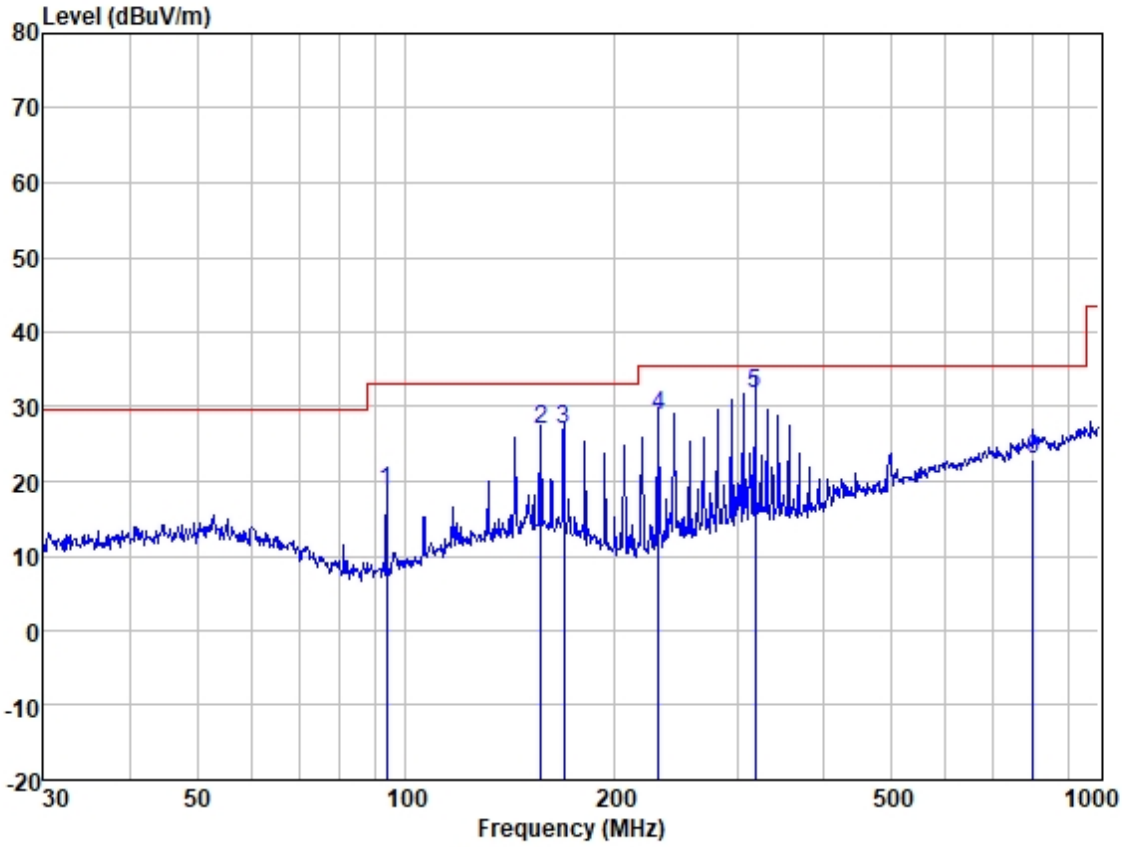
6.2.4 Measurement Procedure and Data

An initial pre-scan was performed in the chamber using the spectrum analyser in peak detection mode. Quasi-peak measurements were conducted based on the peak sweep graph. The EUT was measured by BiConiLog antenna with 2 orthogonal polarities.

Remark: Level= Read Level+ Cable Loss+ Antenna Factor- Preamp Factor

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Test Mode: 00; Polarity: Horizontal



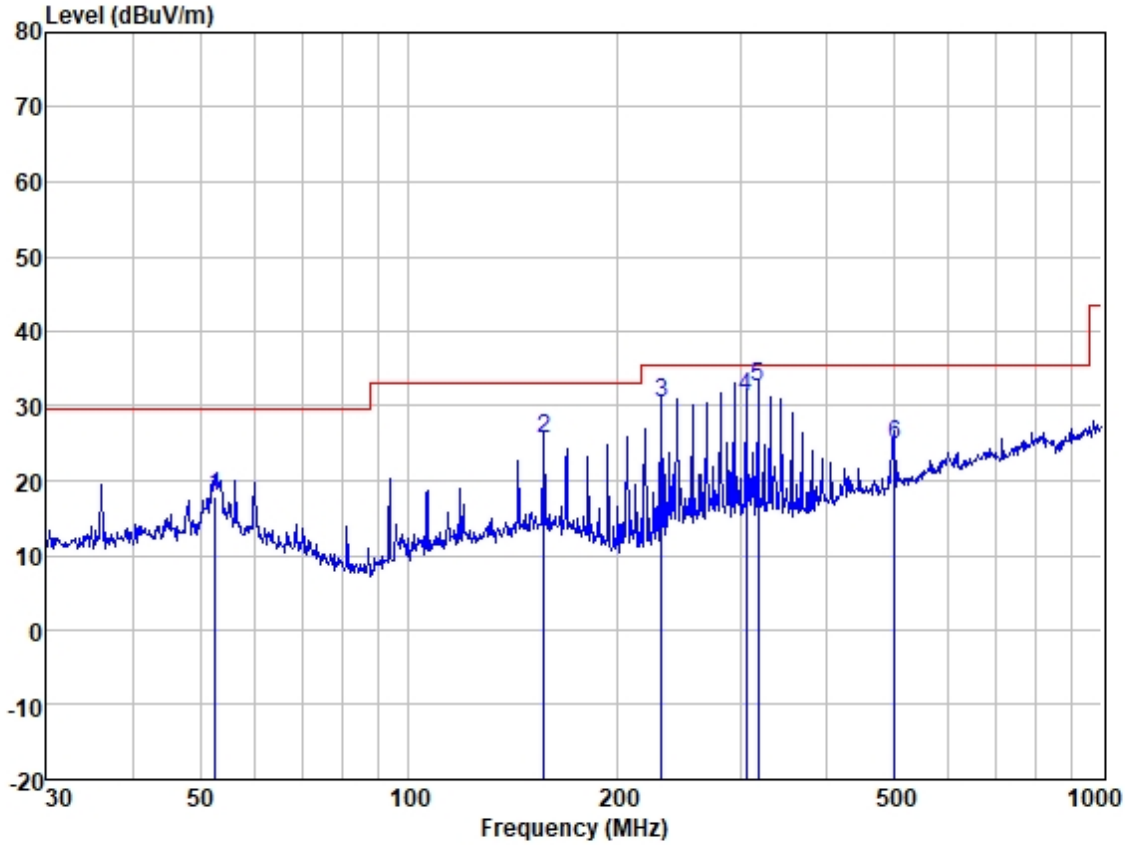
Site : SGS
Job :
Model :
Power :
Test Mode : RT2

	Freq	Read Level	Anterna Factor	Cable Loss	Preamp Factor	Measured Level	Limit Line	Over Limit	Pol/ Phase	Remark
	MHz	dBuV	dB/m	dB	dB	dBuV/m	dBuV/m	dB		
1	93.768	37.85	8.18	0.64	28.10	18.57	33.06	-14.49	HORIZONTAL	QP
2	156.458	40.27	13.61	0.84	27.79	26.93	33.06	-6.13	HORIZONTAL	QP
3	169.005	40.44	13.30	0.88	27.70	26.92	33.06	-6.14	HORIZONTAL	QP
4	231.718	44.28	11.00	1.02	27.56	28.74	35.56	-6.82	HORIZONTAL	QP
5	318.756	43.99	14.01	1.21	27.47	31.74	35.56	-3.82	HORIZONTAL	QP
6	801.786	26.68	22.88	1.93	28.50	22.99	35.56	-12.57	HORIZONTAL	QP

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Test Mode: 00; Polarity: Vertical



Site : SGS
Job :
Model :
Power :
Test Mode : RT2

	Freq	Read Level	Anterna Factor	Cable Loss	Preamp Factor	Measured Level	Limit Line	Over Limit	Pol/ Phase	Remark
	MHz	dBuV	dB/m	dB	dB	dBuV/m	dBuV/m	dB		
1	52.575	31.51	13.93	0.48	28.10	17.82	29.54	-11.72	VERTICAL	QP
2	156.458	38.98	13.61	0.84	27.79	25.64	33.06	-7.42	VERTICAL	QP
3	231.718	45.95	11.00	1.02	27.56	30.41	35.56	-5.15	VERTICAL	QP
4	306.242	43.80	13.65	1.18	27.42	31.21	35.56	-4.35	VERTICAL	QP
5	318.746	44.79	14.01	1.21	27.47	32.54	35.56	-3.02	VERTICAL	QP
6	501.179	34.13	17.76	1.53	28.70	24.72	35.56	-10.84	VERTICAL	QP

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6.3 Radiated Emissions (Above 1GHz)

Test Requirement:	47 CFR Part 15, Subpart B
Test Method:	ANSI C63.4:2014
Limit:	Class B
Above 1GHz	74(dBμV/m) peak, 54(dBμV/m) average
Detector:	Peak for pre-scan (1000kHz resolution bandwidth) 1000M to18000MHz

6.3.1 E.U.T. Operation

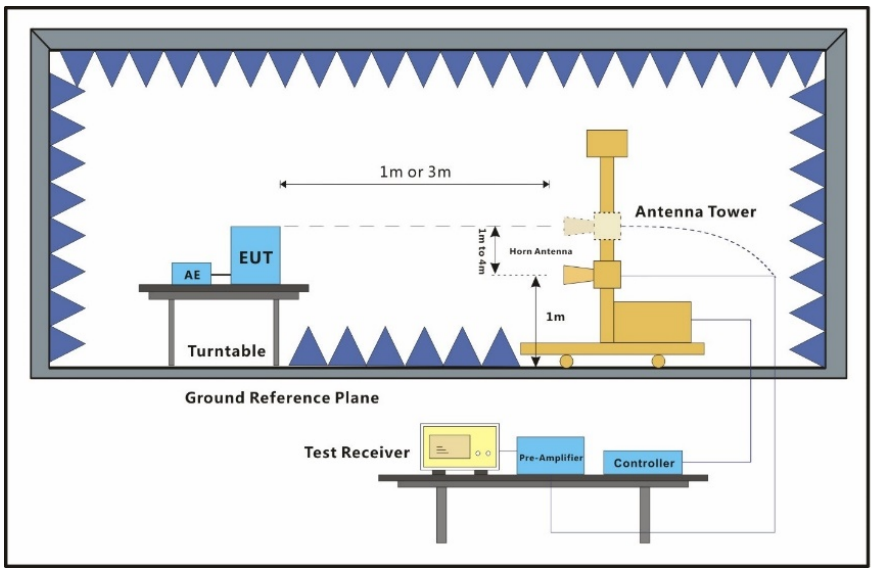
Operating Environment:

Temperature: 25.9 °C Humidity: 58.5 % RH Atmospheric Pressure: 1006 mbar

6.3.2 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Operation(GNSS):Keep the EUT power on and GPS connection through tools.
Pre-scan	01	Idle_Keep the EUT standby.

6.3.3 Test Setup Diagram



6.3.4 Measurement Procedure and Data

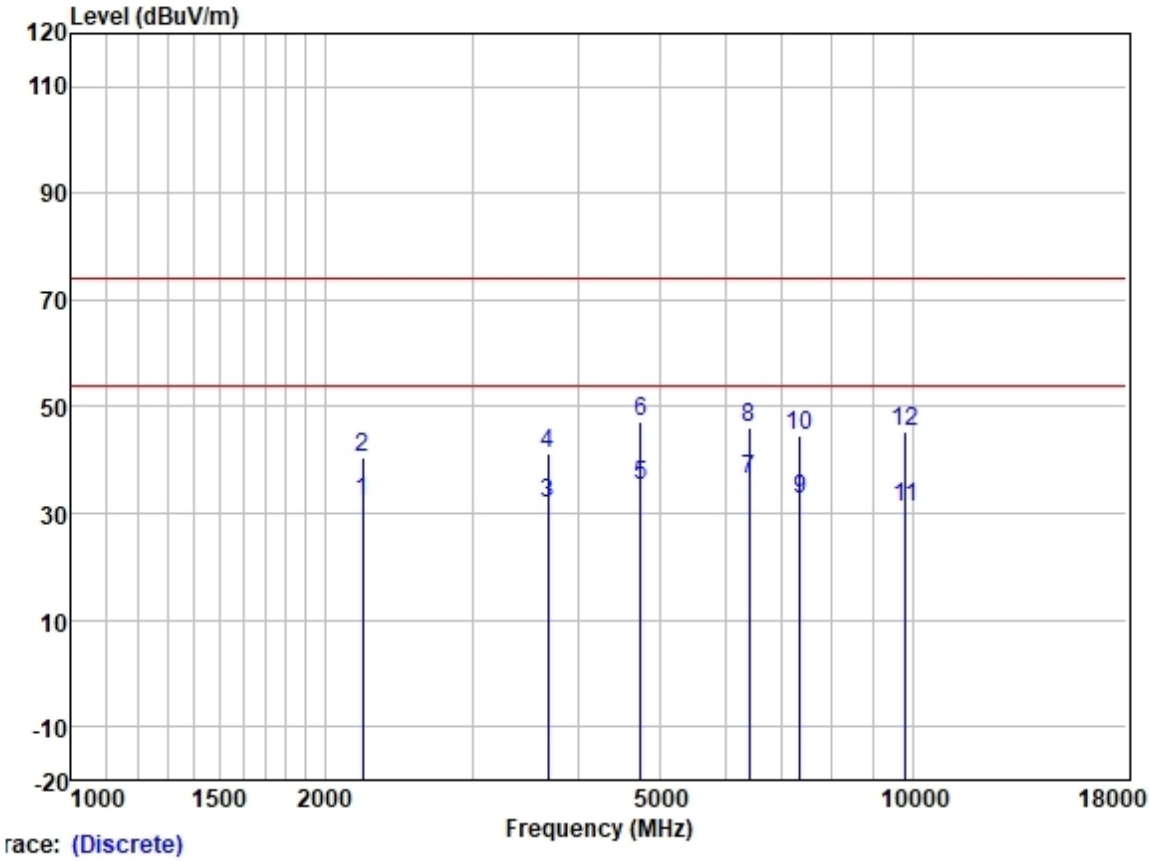
An initial pre-scan was performed in the chamber using the spectrum analyser in peak detection mode. Average measurements were conducted based on the peak sweep graph. The EUT was measured by Horn antenna with 2 orthogonal polarities.

The red line show in graphic is the limit in standard used in this section.

Remark: Level= Read Level+ Cable Loss+ Antenna Factor- Preamp Factor

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Test Mode: 00; Polarity: Horizontal

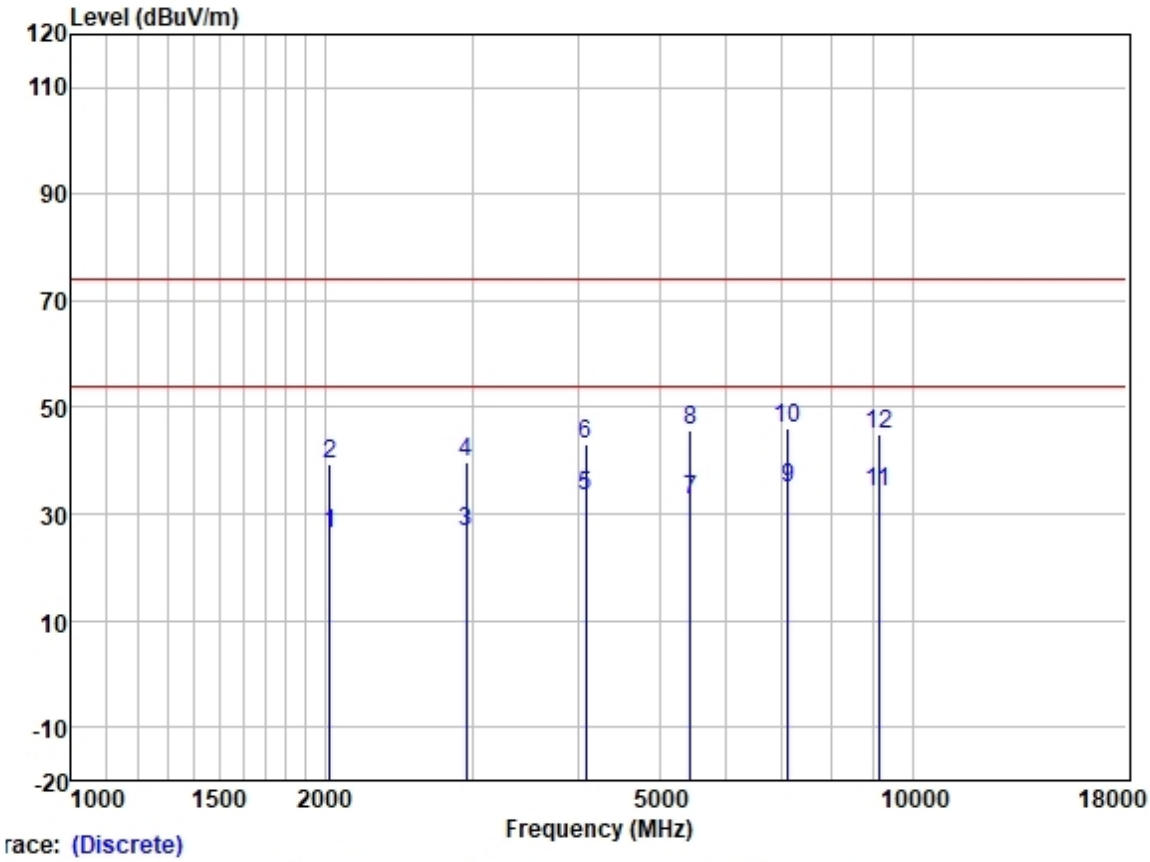


race: (Discrete)

	Freq	Read Level	Antenna Factor	Cable Loss	Preamp Factor	Level	Limit Line	Over Limit	Pol/Phase	Remark
	MHz	dBuV	dB/m	dB	dB	dBuV/m	dBuV/m	dB		
1	2220.523	36.64	27.04	6.28	37.83	32.13	54.00	-21.87	HORIZONTAL	Average
2	2220.523	45.12	27.04	6.28	37.83	40.61	74.00	-33.39	HORIZONTAL	Peak
3	3682.374	34.73	29.55	4.49	36.90	31.87	54.00	-22.13	HORIZONTAL	Average
4	3682.374	44.25	29.55	4.49	36.90	41.39	74.00	-32.61	HORIZONTAL	Peak
5	4748.888	32.60	34.16	5.42	36.86	35.32	54.00	-18.68	HORIZONTAL	Average
6	4748.888	44.57	34.16	5.42	36.86	47.29	74.00	-26.71	HORIZONTAL	Peak
7	6395.654	33.61	33.68	6.20	37.02	36.47	54.00	-17.53	HORIZONTAL	Average
8	6395.654	43.37	33.68	6.20	37.02	46.23	74.00	-27.77	HORIZONTAL	Peak
9	7347.474	27.47	36.12	6.31	37.15	32.75	54.00	-21.25	HORIZONTAL	Average
10	7347.474	39.19	36.12	6.31	37.15	44.47	74.00	-29.53	HORIZONTAL	Peak
11	9809.916	22.20	38.86	7.17	37.01	31.22	54.00	-22.78	HORIZONTAL	Average
12	9809.916	36.25	38.86	7.17	37.01	45.27	74.00	-28.73	HORIZONTAL	Peak

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Test Mode: 00; Polarity: Vertical



race: (Discrete)

	Freq	Read Level	Antenna Factor	Cable Loss	Preamp Factor	Level	Limit Line	Over Limit	Pol/Phase	Remark
	MHz	dBuV	dB/m	dB	dB	dBuV/m	dBuV/m	dB		
1	2030.214	32.09	26.43	5.72	37.89	26.35	54.00	-27.65	VERTICAL	Average
2	2030.214	45.20	26.43	5.72	37.89	39.46	74.00	-34.54	VERTICAL	Peak
3	2947.623	31.51	28.64	3.81	37.41	26.55	54.00	-27.45	VERTICAL	Average
4	2947.623	44.76	28.64	3.81	37.41	39.80	74.00	-34.20	VERTICAL	Peak
5	4086.182	34.00	31.04	5.25	36.80	33.49	54.00	-20.51	VERTICAL	Average
6	4086.182	43.63	31.04	5.25	36.80	43.12	74.00	-30.88	VERTICAL	Peak
7	5439.885	31.23	32.74	5.62	36.96	32.63	54.00	-21.37	VERTICAL	Average
8	5439.885	44.22	32.74	5.62	36.96	45.62	74.00	-28.38	VERTICAL	Peak
9	7117.542	30.11	35.38	6.38	37.12	34.75	54.00	-19.25	VERTICAL	Average
10	7117.542	41.50	35.38	6.38	37.12	46.14	74.00	-27.86	VERTICAL	Peak
11	9126.063	25.87	37.80	7.35	37.07	33.95	54.00	-20.05	VERTICAL	Average
12	9126.063	36.83	37.80	7.35	37.07	44.91	74.00	-29.09	VERTICAL	Peak

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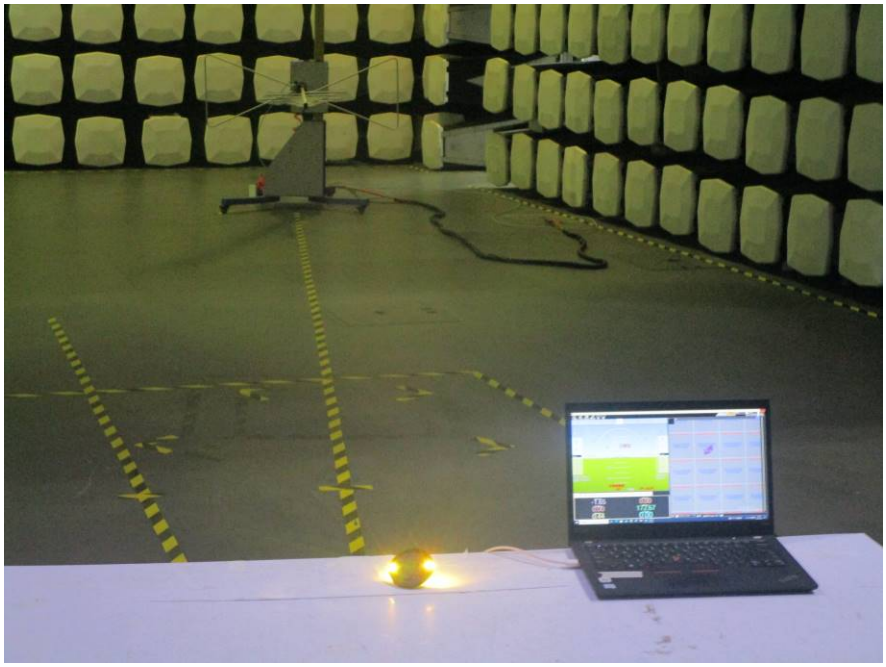
7 Test Setup Photo

Conducted Emissions at Mains Terminals (150kHz-30MHz)

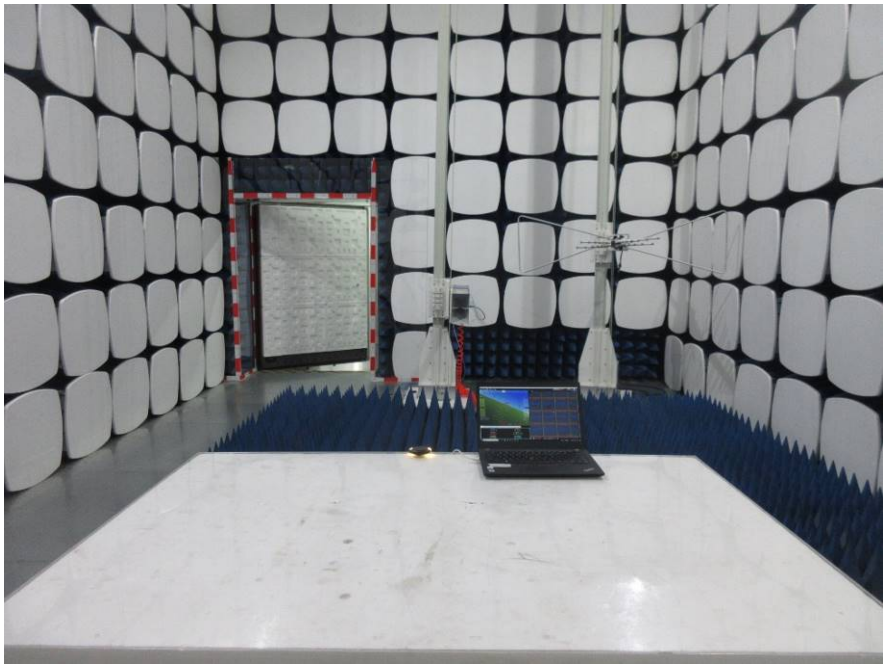


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Radiated Emissions (30MHz-1GHz)

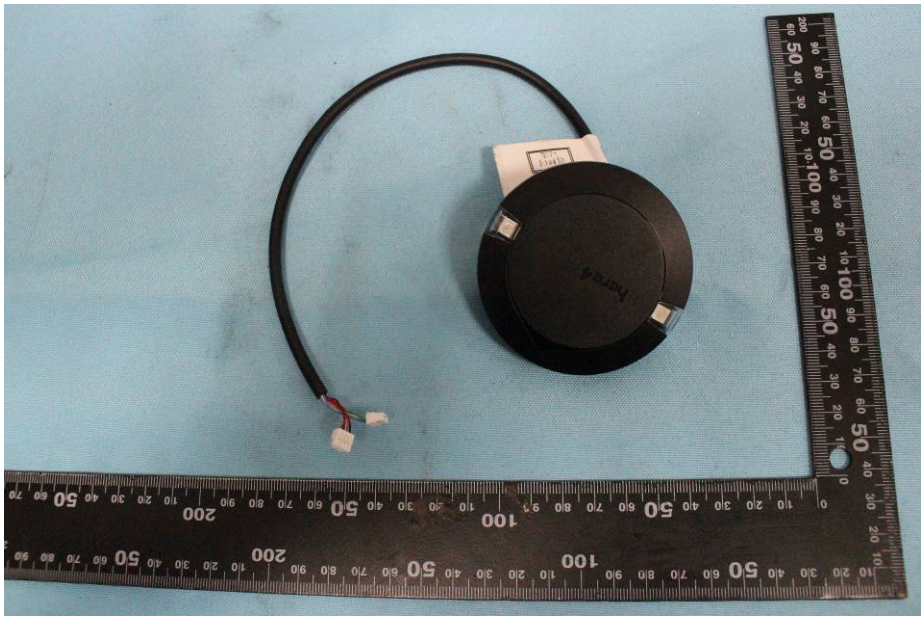


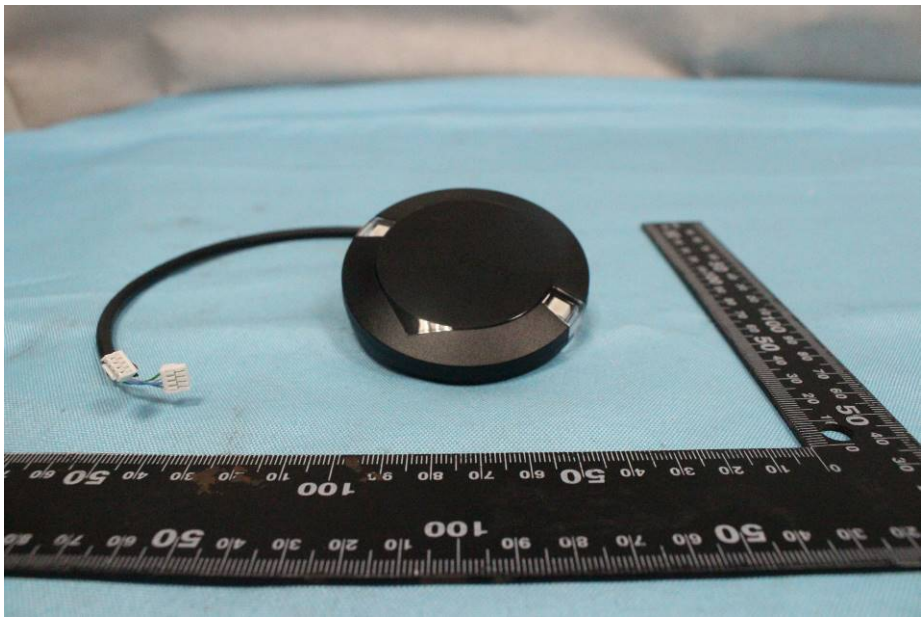
Radiated Emissions (Above 1GHz)



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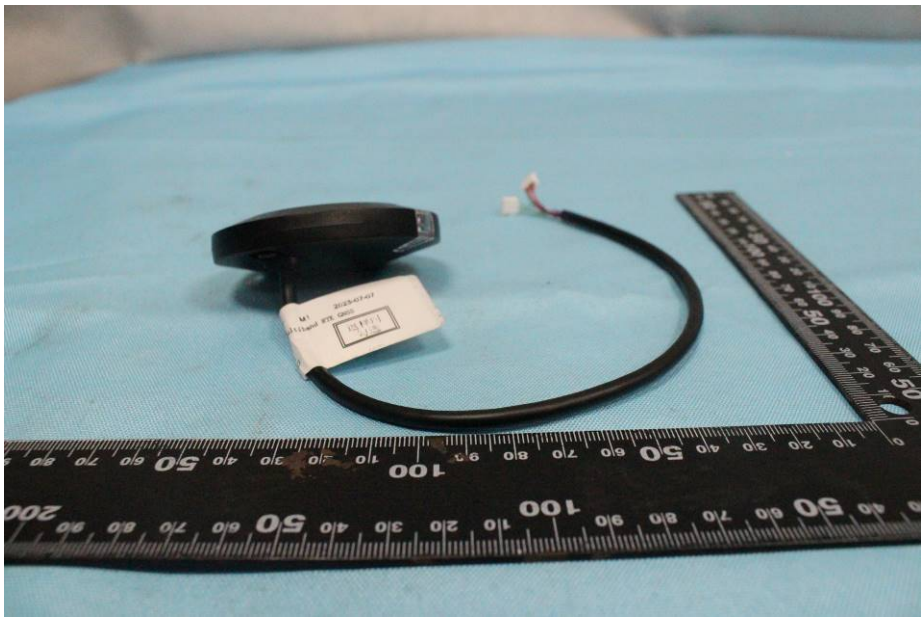
8 EUT Constructional Details (EUT Photos)



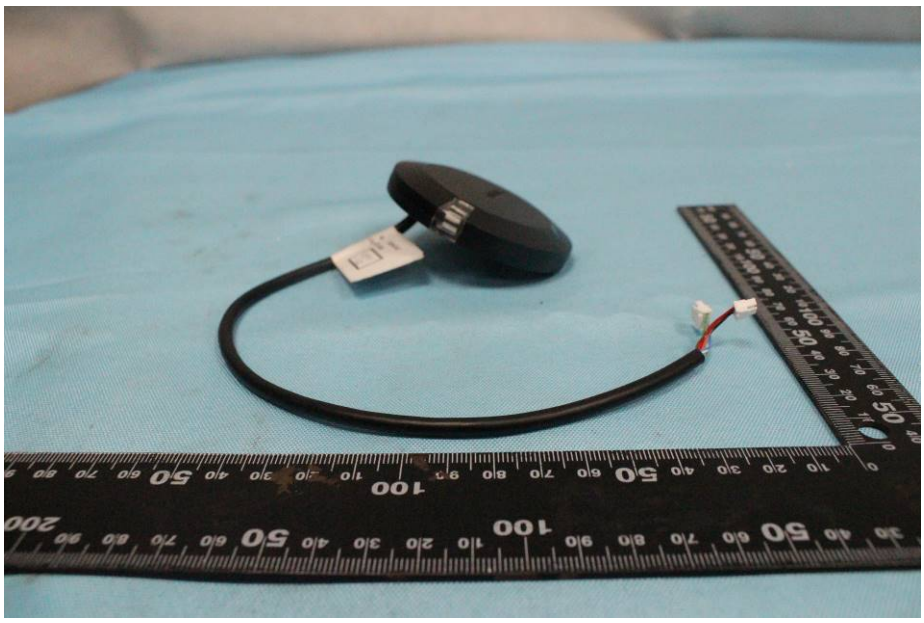


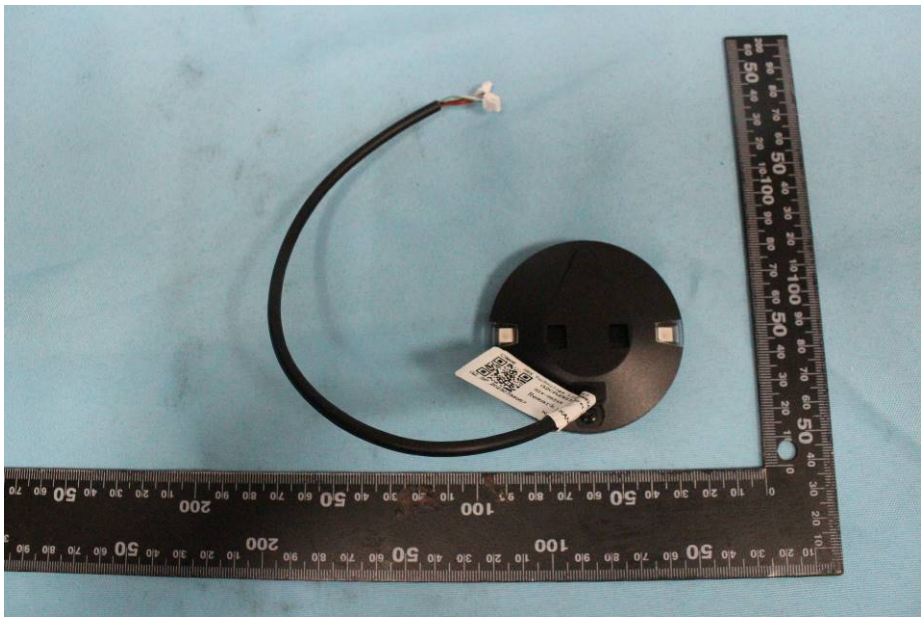
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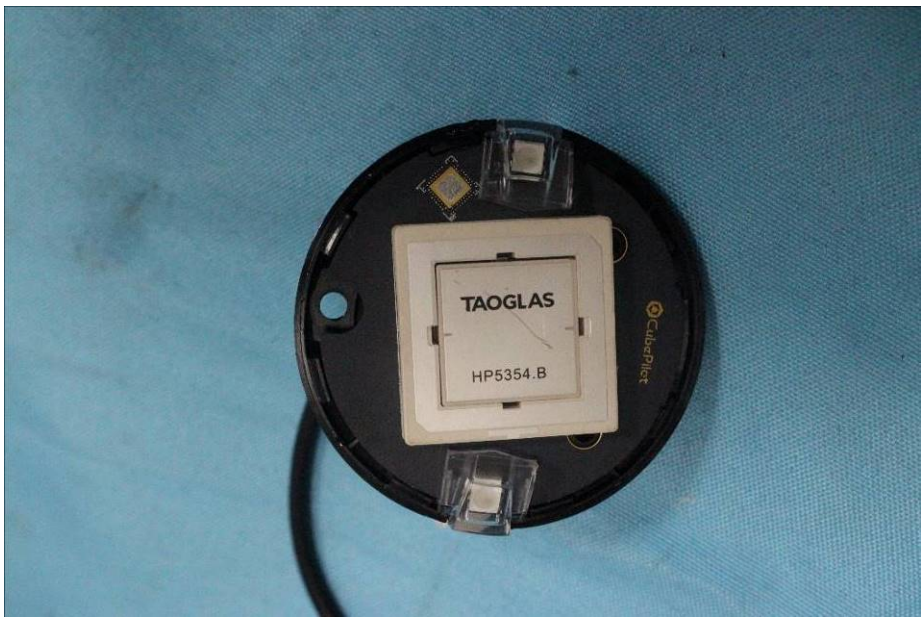


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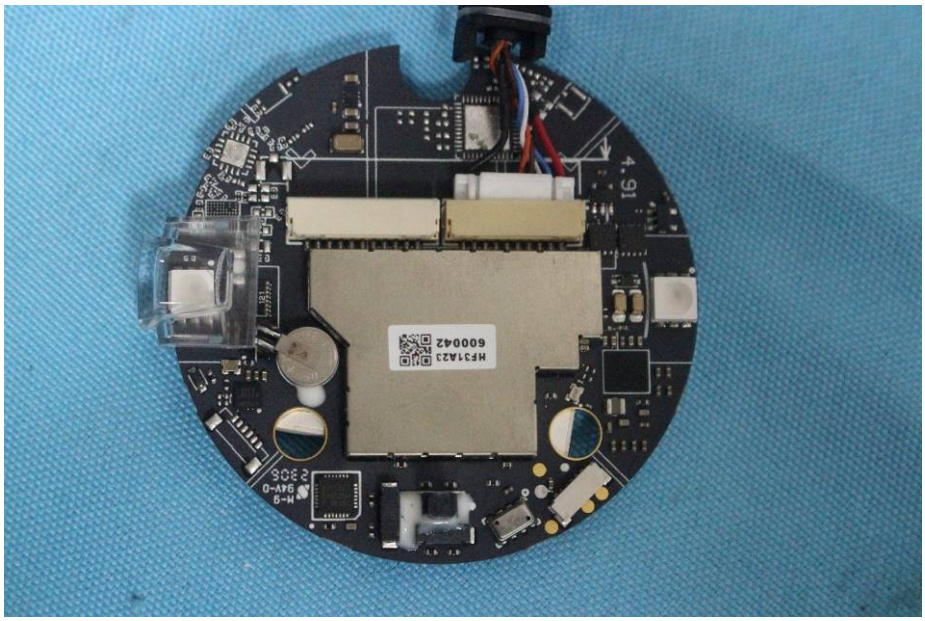


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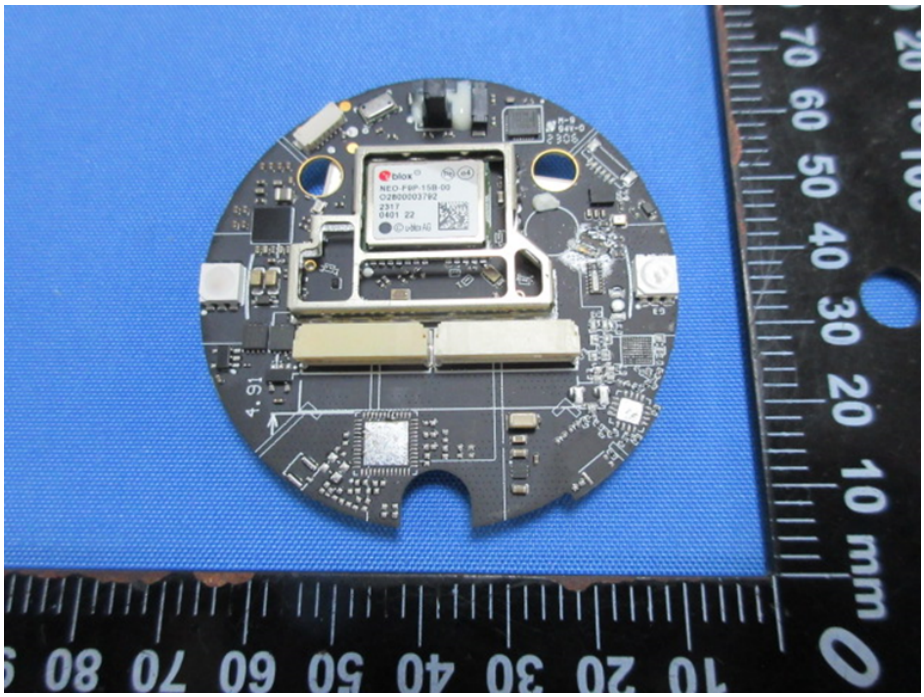




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